

California Housing Price Prediction

Project objective Build a baseline machine-learning model that predicts median house value from California housing-district characteristics, then evaluate its performance on held-out data.

1. Dataset Overview

The project uses the California Housing dataset provided by scikit-learn through `fetch_california_housing(as_frame=True)`. The dataset contains 20,640 observations and eight input features, plus the target variable `MedHouseVal`. The target is expressed in units of \$100,000, so a target value of 2.0 corresponds to approximately \$200,000.

Input features

Feature	Description
MedInc	Median income in the district
HouseAge	Median house age in the district
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	District population
AveOccup	Average household occupancy
Latitude / Longitude	Geographic location
MedHouseVal (target)	

2. Exploratory Data Analysis (EDA)

EDA was used to inspect the dataset, summarize numeric columns, check for missing values, and explore relationships between features and the target. The notebook code uses `df.head()`, `df.info()`, `df.describe()`, and `df.isnull().sum()` for these checks.

Key observations

- **Missing values:** The standard scikit-learn California Housing dataset has no missing values in these columns, which should be verified in the notebook output.
- **Income is strongly associated with value:** The correlation heatmap typically shows a positive correlation of about 0.69 between `MedInc` and `MedHouseVal`. Higher district median income is generally associated with higher median house value.
- **Other individual relationships are weaker:** Features such as `AveRooms` and `HouseAge` usually show much weaker linear correlations with the target.
- **Some predictors are correlated with each other:** `AveRooms` and `AveBedrms` are strongly correlated; latitude and longitude also show a strong relationship in this dataset. This can affect interpretation of individual linear-regression coefficients.
- **Distribution:** The target histogram helps reveal the spread and concentration of district house values. The target is capped at the upper end in the commonly used dataset, which limits the model's ability to represent the most expensive districts

3. Modeling Approach

The target variable is `MedHouseVal`. All eight remaining columns are used as input features (X), while `MedHouseVal` is the output (y). The data is split into training and testing sets with an 80:20 ratio using `train_test_split(test_size=0.2, random_state=42)`. This creates a repeatable split of 16,512 training rows and 4,128 test rows. A `LinearRegression` model from scikit-learn is fitted on the training set. It learns a linear relationship between the input features and the target. The trained model then predicts values for the held-out test set, which were not used to fit the model.

Evaluation metrics

Metric	What it measures	Interpretation
MAE	Mean absolute error	Average absolute prediction error, lower is better
RMSE	Root mean squared error	Penalizes larger errors more than MAE; lower is better
R2	Coefficient of determination	Measures explained variation relative to a mean baseline, closer to 1 is generally better.

4. Results and Interpretation

The notebook reports MAE, RMSE, and R^2 after predicting the test set. These results should be copied from the actual run before submission; this report intentionally does not invent metric values.

Metric	Your notebook result
MAE	Add actual value here
RMSE	Add actual value here
R2	Add actual value here

Diagnostic plots

Actual vs. predicted plot: Points closer to the diagonal reference line indicate more accurate predictions. A wide spread around the line indicates prediction error; systematic departures may reveal patterns the linear model does not capture.

Residual plot: Residuals are calculated as actual minus predicted values. Ideally, they are scattered around zero without a strong curve or funnel shape. Patterns can indicate non-linearity, unequal error variance, or missing explanatory information.

5. Limitations

- Linear Regression assumes an additive linear relationship between predictors and target, which may not fully describe housing prices.
- The dataset represents California districts and should not be treated as a current property valuation tool for individual homes.
- The target values are capped at the upper end in this dataset, so high-value areas may be systematically underpredicted.
- A random train/test split evaluates performance on held-out rows from the same dataset, but does not guarantee performance on other regions or future periods.

6. Improvement Ideas

- **Compare with a baseline:** Evaluate a model that always predicts the training-set mean to establish a simple reference point.
- **Try nonlinear models:** Compare Linear Regression with Random Forest Regressor and Gradient Boosting methods.
- **Use cross-validation:** Estimate performance across multiple folds instead of relying on one train/test split.
- **Feature engineering:** Explore transformations, interactions, and geographic features; inspect outliers and skewed distributions.
- **Tune and validate carefully:** Select models and hyperparameters using training data/cross-validation, then keep the test set for final evaluation.

Conclusion

This project demonstrates a complete introductory regression workflow: data loading, EDA, feature/target selection, train/test splitting, model fitting, prediction, metric evaluation, and diagnostic visualization. Linear Regression provides an interpretable baseline. The next step is to insert the actual MAE, RMSE, and R^2 from the notebook and compare more advanced models to determine whether prediction quality can be improved.